Decisions and acceptance conditions

Requirements trace directly to ten questions and observed source fields.

Five U.S. balancing authorities (CISO, ERCO, ISNE, NYIS and PJM), with publisher operating dates from 1 January 2024 through 31 December 2025. The acquired baseline contains 87,720 authority-hour records over 731 operating dates.

ID	Requirement	Acceptance condition
FR01	Demand profiles and peaks	Q1-Q4 use one demand row per authority/UTC hour and show units, coverage and local-time grouping.
FR02	Forecast evaluation	Q5-Q6 use the same eligible actual/forecast pair set and disclose pair counts.
FR03	Source mix	Q7-Q9 retain source-header semantics and combine facts only after compatible aggregation.
FR04	Source traceability	Every fact maps to a source version and data-record number. Source hashes match the acquisition manifest.
FR05	Data-quality reporting	Q10 distinguishes missing rows, blank cells, invalid numbers and reconciliation flags.
NFR01	Repeatability	A repeat extraction from unchanged cached files yields identical selected CSV hashes and profile counts.
NFR02	Reproducibility	Document Python commands, public URLs, date scope, input hashes and code versions. The bulk download requires no credentials.
NFR03	Maintainability	A newly observed header is reviewed before publication. Exact source field names are recorded in the mapping.
NFR04	Usability	Reports label MW versus MWh, source version, local/UTC time and eligibility. Ratios with no denominator show N/A.

Business owners are represented by electricity planners, operations analysts, energy analysts and the project data steward. Their needs are described in the proposal. The team will validate source facts and analytical interpretations together before implementation.

Population, time and source semantics

These rules govern all reports and form the contract for later ETL.

Population

Use the five selected authorities and filter by DimDate.CalendarDate, which comes from the publisher's Data Date. Each source interval is one hour ending at UTCHourEnd. Use UTC for uniqueness and simultaneous cross-area comparisons. Use source local hour labels and operating dates for local profiles. Report the source snapshot and eligible count beside every result.

Power and energy

The acquired bulk headers specify MW. Store those reported hourly-average power values in MW. Compute interval energy as MW multiplied by one hour and report time totals in MWh. Average or maximize MW across hours. Sum power across distinct authorities only at aligned UTC instants.

Local and UTC time

Parse UTC Time at End of Hour with an explicit UTC offset. Parse Local Time at End of Hour as the source wall-clock label. Store their difference as UTCOffsetMinutes. The two autumn repetitions per authority remain distinct because their UTC instants differ. A midnight end label can belong to the previous operating date, so DateKey comes from Data Date.

Fuel categories

Preserve exact reported fuel headers as category natural keys. The first 2024 file has nine fuel headers, while the next three have sixteen. Across the baseline there are nineteen distinct header meanings and sixteen with numeric observations. Roll up related solar and wind categories only through the documented mapping. The legacy combined hydro/pumped-storage category remains explicit.

Missing observations

Keep blank measures as null. Keep verified zero as zero, while excluding nonpositive actual demand from forecast scoring. A missing fuel column means the category is absent from that file's schema; a blank cell means no value was supplied. Neither proves zero generation. Unresolvable identity or timestamp rows enter the exception queue.

Combining facts

FactHourlyDemand is unique on authority and UTC hour. FactHourlyGeneration adds energy source to that grain. Aggregate generation to authority and UTC hour before joining to demand. For a group comparison, aggregate each fact independently to the same dimensions and then combine. One shared authority-hour with eight fuel categories still has one demand observation.

Demand and forecast calculations

Zero-denominator rates display N/A. A missing value is retained as a missing value.

Metric	Definition and eligibility
M01 Observed demand energy	Sum ActualDemandMW x 1 hour over numeric, nonnegative actual observations. State coverage. Unit MWh. This is observed energy when hours have missing measures.
M02 Mean hourly demand	Sum eligible ActualDemandMW / number of eligible observations. Numeric nonnegative observations qualify. Group by local hour, operating date attributes and authority. Unit MW.
M03 Peak hourly demand	Maximum eligible ActualDemandMW within each authority and operating-month slice. Return every tied UTC instant with its local label. Unit MW.
M04 Forecast error and bias	Eligible pairs have numeric ActualDemandMW > 0 and ForecastDemandMW >= 0. Error = actual minus forecast. Bias = sum(error) / eligible pair count. Positive bias means underforecasting. Unit MW.
M05 MAE	Sum(abs(actual - forecast)) / eligible pair count, using the M04 population. Unit MW. Return N/A when no pair qualifies.
M06 WAPE	100 x sum(abs(actual - forecast)) / sum(actual), using exactly the M04 population. Return N/A for zero denominator. Aggregate from the sums, never by averaging area percentages.
M07 Underforecast frequency	100 x count(actual > forecast) / eligible pair count. Equal values remain in the denominator. Use M04 eligibility.

Published forecasts

The comparison uses the forecast values present in the downloaded historical file. It does not claim to reconstruct the exact forecast version that was available before the operating day. Source snapshot identifiers make later changes traceable.

Generation and coverage calculations

Every share or cross-fact comparison uses matched authority-hour populations.

Metric	Definition and eligibility
M08 Reported net generation energy	Sum numeric signed GenerationMW x 1 hour by original source category or mapped SourceGroup. Retain negative values and indicate coverage. Unit MWh.
M09 Wind and solar category share	100 x sum(wind and solar GenerationMW x 1 hour) / sum(TotalGenerationMW x 1 hour). Use matched authority-hours with observed wind and solar categories, positive total generation and an available-fuel residual within max(10 MW, 1% of abs(total)). Show retained/excluded hours and category versions. Missing components are never silently set to zero.
M10 High-demand hours	Within each authority and operating month, compute the 90th percentile of numeric positive actual demand. Flag actual >= percentile, retaining ties. Compare source mix in that set with remaining eligible hours after generation is summarized at authority-hour grain.
M11 Missing / invalid measure rate	100 x affected expected authority-hours / expected authority-hours in the selected coverage window. Diagnose absent source rows separately from present rows with null or invalid measures. The frozen baseline's expected grid is 87,720 hours.
M12 Available-fuel reconciliation residual	Sum numeric reported fuel values at authority-hour minus TotalGenerationMW. Compare only when the total and at least one fuel value exist. Flag abs(residual) > max(10 MW, 1% of abs(total)). The threshold is a project diagnostic, not an EIA rule. A small residual does not prove every fuel category was reported.

Keep immutable raw extracts and their hashes. A changed bulk file creates a new DimSourceSnapshot member. A later load updates measures at the same business grain and preserves fact surrogate IDs. Replacement of a source partition must also remove fuel facts for values withdrawn by the publisher, or mark them absent through a reviewed policy. A latest-value fact cannot reproduce an old publication without its archived source snapshot.

Controls before a report is published

Measured source exceptions are retained for investigation and reflected in reporting coverage.

Control	Rule
Identity	Require the selected BA code, valid UTC hour and a unique authority/hour pair per source version. Hold contradictory duplicates.
Timestamps	Preserve UTC identity and source local labels. Validate source operating dates, whole-hour instants and observed offsets. Never deduplicate by local clock alone.
Missing values	Retain source blanks as null, distinguish absent headers and publish coverage. No zero-fill for unreported fuel values.
Numeric conversion	Parse signed finite decimals. Quarantine malformed numbers with source locator. Retain zero and negative source values for review.
Forecast eligibility	Score only actual > 0 and forecast >= 0 with both numeric. Count excluded pairs.
Generation reconciliation	Compare the available reported source sum with reported total. Flag beyond max(10 MW, 1% of abs(total)). Preserve discrepancies for source investigation.
Schema change	Use the exact source-header mapping and file schema. Keep old combined and new split categories distinct, including all-null categories.
Referential integrity	Load parent dimensions before facts. Accepted facts must have zero unresolved foreign keys. A malformed time/code stays in staging exceptions.
Revisions	Keep immutable raw snapshots. Reprocess changed source partitions atomically. Reconcile insert/update/removal counts before committing load watermarks.
Safe aggregation	Check that adding fuel detail never increases demand totals or demand observation counts. Recompute rates from numerator and denominator sums.

No user-level personal data is required. Reproducibility depends on immutable raw evidence, a documented mapping release and committed load boundaries. Phase 1 acceptance covers design consistency and source feasibility; database row counts and ETL execution checks will be measured in Phase 2.